KING SALMAN INTERNATIONAL UNIVERSITY

Field of Technological Industries

All Program

Course : Mathematics 1 (MAT 111)

Lecture 5: Derivatives (Basic Rules)

Dr : Ahmed Ali

Course contents

Calculus part

- Functions
- Limits
- Continuity
- Derivatives
- Applications of differentiation

Linear Algebra part

- Systems of Linear Equations
- Theory of Matrices
- Applications in Systems of Equations

textbook: R. Smith, R. Minton, z. Rafhi, Calculus: Early Transcendental Functions, 5th Edition, McGraw-Hill Education.

DIFFERENTATION

What is meant by differentiation?

It's the slope of the tangent for the curve $y = F(x)$ at a certain point.

Say, we have $y = F(x)$

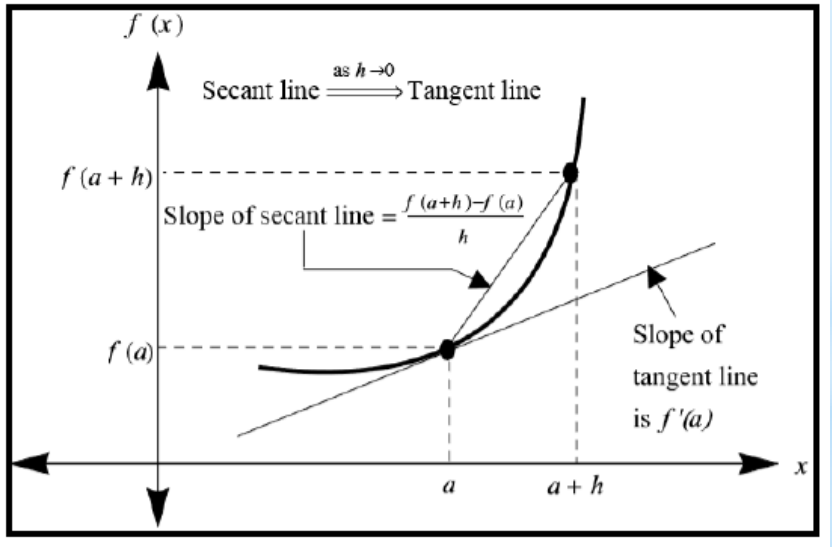
$$y' = \frac{dy}{dx} = \tan\theta = \lim_{\Delta x \to 0} \frac{\Delta y}{\Delta x} = \lim_{\Delta x \to 0} \frac{F(x+\Delta x) - F(x)}{\Delta x}$$

Say, $y = C$

$$y' = \frac{dy}{dx} = \lim_{\Delta x \to 0} \frac{C - C}{\Delta x} = 0$$

Say, $y = x$

$$y' = \frac{dy}{dx} = \lim_{\Delta x \to 0} \frac{(x+\Delta x) - x}{\Delta x} = 1$$



Say, $y = x^2$

$$y' = \frac{dy}{dx} = \lim_{\Delta x \to 0} \frac{(x+\Delta x)^2 - x^2}{\Delta x} = \lim_{\Delta x \to 0} \frac{x^2 + 2x\Delta x + (\Delta x)^2 - x^2}{\Delta x}$$

$$= \lim_{\Delta x \to 0} \frac{\Delta x(2x + \Delta x)}{\Delta x} = \lim_{\Delta x \to 0} (2x + \Delta x) = 2x$$

Basic Rules:

(1) $y = C \Longrightarrow y' = \frac{dy}{dx} = 0$, i.e. $y = 2 \Longrightarrow y' = 0$

(2) $y = x^n \Longrightarrow y' = \frac{dy}{dx} = nx^{n-1}$

Example:

$y = x^5 \Longrightarrow y' = 5x^4$

(3) $y = C\,F(x) \Longrightarrow y' = \frac{dy}{dx} = C.F'(x)$

Example:

$y = 3x^4$, then $y' = 3 * 4x^3 = 12x^3$

(4) $y = F(x) \pm g(x)$, then $y' = F'(x) \pm g'(x)$

Example:

$$y = 3x^5 - 4x^3 + 3x - 6$$

Solution:

$$y' = 15x^4 - 12x^2 + 3$$

(5) ***The product rule:*** $y = u.v$

$$y' = u.v' + v.u'$$

Example:

$$y = (3x^4 - 2x + 4)(2x^2 + 6)$$

Solution:

$$y' = (3x^4 - 2x + 4)(4x) + (2x^2 + 6)(12x^3 - 2)$$

(6) ***The quotient rule:*** $y = \frac{u}{v}$

$$y' = \frac{v.u' - u.v'}{v^2}$$

Example:

$$y = \frac{(3x^4 - 3x + 2)}{(x^7 + 3x^2)}$$

Solution:

$$y' = \frac{(x^7 + 3x^2)(12x^3 - 3) - (3x^4 - 3x + 2)(7x^6 + 6x)}{(x^7 + 3x^2)^2}$$

In general:

(7)

$$y = u^n$$

$$y = nu^{n-1}.u'$$

Example:

$$y = (3x^{-4} - 3x^3 + 2)^5$$

Solution:

$$y' = 5(3x^{-4} - 3x^3 + 2)^4(-12x^{-5} - 9x^2)$$

(8) $y = \sqrt{x}$

which can be written as: $y = x^{\frac{1}{2}}$

$$y' = \frac{1}{2}x^{-\frac{1}{2}} = \frac{1}{2}\frac{1}{x^{\frac{1}{2}}} = \frac{1}{2\sqrt{x}}$$

Example:

$$y = 3\sqrt{x}$$

Solution:

$$y' = 3.\frac{1}{2\sqrt{x}} = \frac{3}{2\sqrt{x}}$$

Example:

$$y = 5\sqrt{x}$$

Solution:

$$y' = \frac{5}{2\sqrt{x}}$$

Example:

$$y = \sqrt[3]{x}$$

Solution:

$$y = x^{\frac{1}{3}}$$

$$y' = \frac{1}{3}x^{-\frac{2}{3}}$$

Example:

$$y = \sqrt[4]{x}$$

Solution:

$$y = x^{\frac{1}{4}}$$

$$y' = \frac{1}{4}x^{-\frac{3}{4}}$$

(9)

$$y = \sqrt{u}$$

$$y' = \frac{1}{2\sqrt{u}}.u'$$

Example:

$$y = \sqrt{x^2 - 4x + 3}$$

Solution:

$$y' = \frac{1}{2\sqrt{x^2 - 4x + 3}}.(2x - 4)$$

Example:

$$y = \frac{1}{x}$$

Solution:

$$y = x^{-1}$$

$$y' = -x^{-2} = -\frac{1}{x^2}$$

No. 1 $h(x)=\frac{10}{\sqrt[3]{x}}-2x+\pi$

Solution:

$$h(x)=10x^{-\frac{1}{3}}-2x+\pi$$

$$h'(x)=10\left(-\frac{1}{3}\right)x^{-\frac{4}{3}}-2=-\frac{10}{3}x^{-\frac{4}{3}}-2$$

Higher order derivatives:

Say,

$$y=5x^4-2x^3+4x-6$$

First order derivative: $y'=\frac{dy}{dx}=20x^3-6x^2+4$

Second order derivative: $y''=\frac{d^2y}{dx^2}=60x^2-12x$

Third order derivative: $y'''=\frac{d^3y}{dx^3}=120x-12$

Sheet (1)

A. Find $\frac{dy}{dx}$ for each of the following functions

No.1 $y = x^4 - 3x^{-2} + 15x + 10$

Solution:

$$y' = 4x^3 + 6x^{-3} + 15$$

No.2 $y = (\sqrt{x} - 1)^7$

Solution:

$$y' = 7(\sqrt{x} - 1)^6 \left(\frac{1}{2\sqrt{x}}\right)$$

No.3 $y = \frac{1}{(x^6 - 2)^5}$

Solution:

$$y = (x^6 - 2)^{-5}$$

$$y = -5(x^6 - 2)^{-6}(6x^5) = -30x^5(x^6 - 2)^{-6}$$

No.4 $y=(x^3-1)^5(2+3x^{-4})^7$

Solution:

$$y'=(x^3-1)^5.7(2+3x^{-4})^6(-12x^{-5})+(2+3x^{-4})^7.5(x^3-1)^4(3x^2)$$

No.5 $y=\dfrac{x^3-1}{x^3+1}$

Solution:

$$y'=\frac{(x^3+1)(3x^2)-(x^3-1)(3x^2)}{(x^3+1)^2}$$

No.6 $y=\left(\dfrac{x^2-3}{x^{-4}+2}\right)^{\frac{4}{3}}$

Solution:

$$y'=\frac{4}{3}\left(\frac{x^2-3}{x^{-4}+2}\right)^{\frac{1}{3}}\frac{(x^{-4}+2)(2x)-(x^2-3)(-4x^{-5})}{(x^{-4}+2)^2}$$

B. Find $\frac{d^2y}{dx^2}$ for each of the following functions

No.1 $y = x^7 - \frac{2}{x^3} + x^{-5} + 16x + 5$

Solution:

$$y = x^7 - 2x^{-3} + x^{-5} + 16x + 5$$
$$y' = 7x^6 + 6x^{-4} - 5x^{-6} + 16$$
$$y'' = 42x^5 - 24x^{-5} + 30x^{-7}$$

No.2 $y = (2 - x^3)^8$

Solution:

$$y' = 8(2 - x^3)^7(-3x^2) = (-24x^2)(2 - x^3)^7$$
$$y'' = (-24x^2).7(2 - x^3)^6(-3x^2) + (2 - x^3)^7(-48x)$$

C. If $y = (x + \sqrt{x^2 - 1})^4$, Show that $\frac{dy}{dx} = \frac{4y}{\sqrt{x^2 - 1}}$

Solution:

$$y' = 4\left(x + \sqrt{x^2 - 1}\right)^3\left(1 + \frac{1}{2\sqrt{x^2 - 1}}.2x\right)$$

$$y' = 4\left(x + \sqrt{x^2 - 1}\right)^3\left(1 + \frac{x}{\sqrt{x^2 - 1}}\right)$$

$$y' = 4\left(x + \sqrt{x^2 - 1}\right)^3\left(\frac{\sqrt{x^2 - 1} + x}{\sqrt{x^2 - 1}}\right)$$

$$y' = 4\left(\frac{(x + \sqrt{x^2 - 1})^4}{\sqrt{x^2 - 1}}\right) = \frac{4y}{\sqrt{x^2 - 1}}$$

KING SALMAN INTERNATIONAL UNIVERSITY

THANK YOU

16949 www.ksiu.edu.eg info@ksiu.edu.eg

El Tur Campus : Sahely Road Ras suder Campus : Obour Street Sharm El Sheikh Campus : El Nour District